

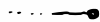
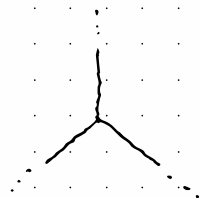
10 March 2022

More Low Dimensional Examples (Nadler)

Last Time. $\dim S^*X = 3$, $\dim \Lambda^\infty = 1$

Λ^∞

—



μSh_{Λ^∞}

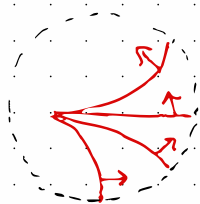
$\text{Mod}(k)$ noncanonically

Δ 's

totally acyclic
n-cycles

0

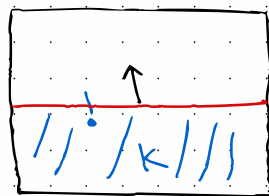
> Choose emb. of Λ^∞ so that front looks like



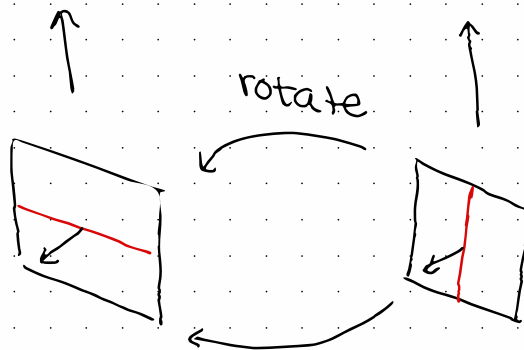
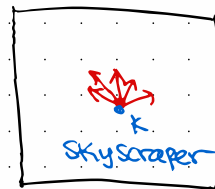
$$\mu Sh_{\Lambda^\infty}|_x \approx Sh_{\Lambda \cup B}(B)^{\Gamma=0}$$

To discuss.

- > non-canonical
- > Invariance



generator



matching of μSh_Λ differs by a shift.

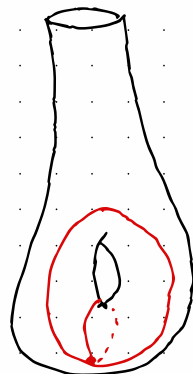
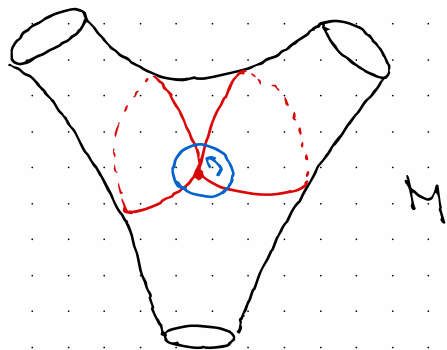
End goal. Given a Lagrangian graph $\Lambda \subset M$ symplectic surface, want to assign a Sheaf of Categories μSh_Λ .

> If M is exact ($\omega = d\alpha$) and Λ is exact (roughly $\alpha|_\Lambda = df$), then Λ lifts to a Legendrian in the Contactification $M \times \mathbb{R}$.

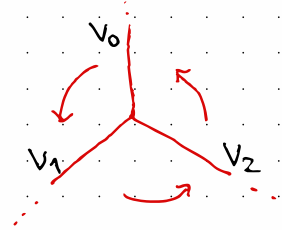
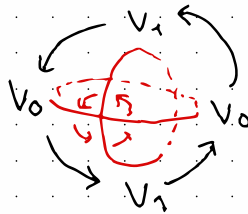
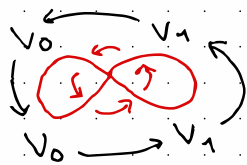
$$- \mathbb{R}^3 = \mathbb{R}_x \times \mathbb{R}_z \times \mathbb{R}_y = \underbrace{J^1(\mathbb{R}_x)}_{1\text{-jet space}} \subset S^*(\mathbb{R}_x \times \mathbb{R}_z)$$

$dz - ydx$ Contact form

$$\begin{array}{ccc} \mathbb{R}_x \times \mathbb{R}_y & \xleftarrow{\text{leg}} \mathbb{R}^3 & \xrightarrow{\text{front}} \mathbb{R}_x \times \mathbb{R}_z \\ \text{SII} & & \\ T^*\mathbb{R}_x & & \end{array}$$



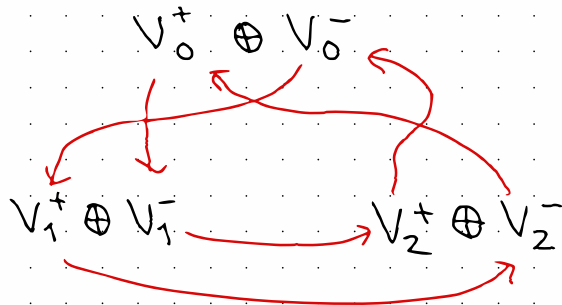
Same graphs
but different
cyclic orderings
 μSh_{∞} will depend on
the cyclic ordering



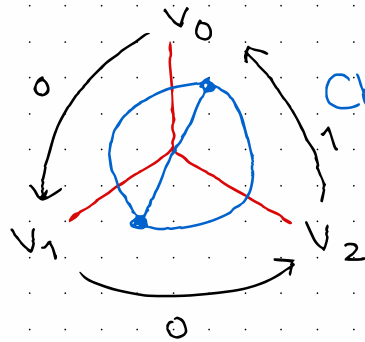
Something we've been sweeping under the rug. One of these maps needs to be of degree 1!

1st correction. Work with $\mathbb{Z}/2$ -graded dg-categories and maps in cycle have degree 1

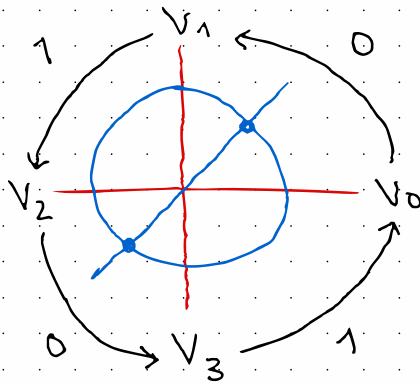
$$A \xrightarrow{[1]} B[1] \xrightarrow{[1]} \text{Cone}(f)[2] \xrightarrow{[1]} A[2] \simeq A$$



2nd Correction. Introduce extra Structure



Choice of a line field in $TM \cong T^*M$



> This is called a 'bicanonical trivialization'

$$\tau: M \xrightarrow{\text{triv}} (T^*_\mathbb{C} M)^{\otimes 2} \Leftrightarrow \pm \sqrt{2} \cdot M \rightarrow T^*M$$



Note. In the cosphere bundle, there is a canonical line field given by the vertical polarization of T^*M .

